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import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import
classification_report, accuracy_score, confusion_matrix
from sklearn.preprocessing import LabelEncoder
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
#loading the dataset
df=pd.read_csv('/content/career_change_prediction_dataset.csv')
#information of file
df.info()
df.head(10)
#determing the null values
print("\nthe missing values:\n")
missing_value=df.isnull().sum()
print(missing_value[missing_value>0])
#handling the missing values
imputer=SimpleImputer(strategy='most_frequent')
df=pd.DataFrame(imputer.fit_transform(df),columns=df.columns)
print("\nthe missing values after handling:\n")
print(df.head(10))
print(df.isnull().sum())
#handling categorical values
label_encoder=LabelEncoder()
categorical_col=df.select_dtypes(include=['object']).columns
for col in categorical_col:
    df[col]=label_encoder.fit_transform(df[col])
print("after handling categorical values:")
print(df.head(10))
#describing
df.describe()
#train test split operation
x=df.drop('Likely to Change Occupation',axis=1)
y=df['Likely to Change Occupation']
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_state=42)
x_train.shape,x_test.shape
y_train.shape,y_test.shape
#training the random forest classifier
model=RandomForestClassifier(n_estimators=100,random_state=42)
model.fit(x_train,y_train)
#making predictions
y_pred=model.predict(x_test)
#classification report

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print("\nclassification report:\n")
print(classification_report(y_test,y_pred))
#accuracy score
print("\naccuracy score:\n")
accuracy=accuracy_score(y_test,y_pred)
print(accuracy)
#confusion matrix
print("\nconfusion matrix:\n")
confmatrix=confusion_matrix(y_test,y_pred)
sns.heatmap(confmatrix,annot=True,fmt='d',cmap='Blues',xticklabels=['stay','switch'],yticklabels=['stay','switch'])
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title('Confusion Matrix')
plt.show()
#pca using built in function
A=df.drop('Likely to Change Occupation',axis=1).values
pca=PCA(n_components=2)
principalComponents=pca.fit_transform(A)
principalDf=pd.DataFrame(data=principalComponents,columns=['principal component 1','principal component 2'])
finaldf=pd.concat([principalDf,df[['Likely to Change Occupation']]],axis=1)
print(finaldf.head())
#graph
fig = plt.figure(figsize = (8,8))
ax = fig.add_subplot(1,1,1)
ax.set_xlabel('Principal Component 1', fontsize = 15)
ax.set_ylabel('Principal Component 2', fontsize = 15)
ax.set_title('2 component PCA', fontsize = 20)
targets = [0,1]
colors = ['r', 'g', 'b']
for target, color in zip(targets,colors):
    indicesToKeep = finaldf['Likely to Change Occupation'] == target
    ax.scatter(finaldf.loc[indicesToKeep, 'principal component 1'],
finaldf.loc[indicesToKeep, 'principal component 2'], c = color, s = 50)
ax.legend(targets)
ax.grid()

```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 38444 entries, 0 to 38443
Data columns (total 23 columns):

```

#	Column	Non-Null Count	Dtype
0	Field of Study	38444 non-null	object
1	Current Occupation	38444 non-null	object
2	Age	38444 non-null	int64

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3 Gender 38444 non-null object
4 Years of Experience 38444 non-null int64
5 Education Level 38444 non-null object
6 Industry Growth Rate 38444 non-null object
7 Job Satisfaction 38444 non-null int64
8 Work-Life Balance 38444 non-null int64
9 Job Opportunities 38444 non-null int64
10 Salary 38444 non-null int64
11 Job Security 38444 non-null int64
12 Career Change Interest 38444 non-null int64
13 Skills Gap 38444 non-null int64
14 Family Influence 28812 non-null object
15 Mentorship Available 38444 non-null int64
16 Certifications 38444 non-null int64
17 Freelancing Experience 38444 non-null int64
18 Geographic Mobility 38444 non-null int64
19 Professional Networks 38444 non-null int64
20 Career Change Events 38444 non-null int64
21 Technology Adoption 38444 non-null int64
22 Likely to Change Occupation 38444 non-null int64
dtypes: int64(17), object(6)
memory usage: 6.7+ MB

```

the missing values:

```

Family Influence    9632
dtype: int64

```

the missing values after handling:

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      Field of Study  Current Occupation  Age  Gender  Years of
Experience \
0      Medicine    Business Analyst  48    Male
7
1      Education    Economist  44    Male
26
2      Education    Biologist  21    Female
27
3      Education    Business Analyst  33    Male
14
4      Arts    Doctor  28    Female
0
5      Medicine    Lawyer  56    Male
39
6      Medicine  Software Developer  51    Female
29
7      Medicine    Artist  58    Female
32
8  Computer Science  Software Developer  31    Female
38

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9	Business	Doctor	30	Female
16				
	Education Level	Industry Growth Rate	Job Satisfaction	Work-Life Balance \
0	High School	High	7	
10				
1	Master's	Low	10	
3				
2	Master's	Low	8	
3				
3	PhD	Medium	7	
9				
4	PhD	Low	3	
1				
5	Bachelor's	Low	3	
1				
6	High School	Low	7	
4				
7	High School	High	9	
4				
8	Bachelor's	Low	1	
5				
9	Master's	Medium	8	
5				
	Job Opportunities Available \	... Skills Gap	Family Influence	Mentorship
0	83	...	8	High
0				
1	55	...	3	Medium
0				
2	78	...	4	Low
0				
3	62	...	2	Medium
1				
4	8	...	5	Low
0				
5	99	...	10	Medium
1				
6	13	...	6	High
1				
7	77	...	8	Medium
0				
8	56	...	6	Medium
0				
9	98	...	6	High
0				
	Certifications	Freelancing Experience	Geographic Mobility	\

0	0	0	1
1	0	1	1
2	0	0	0
3	0	0	0
4	0	1	0
5	0	0	0
6	1	0	1
7	0	0	0
8	1	0	0
9	0	0	0

	Professional Networks	Career Change Events	Technology Adoption	\
0	2	0	1	
1	2	1	9	
2	2	1	2	
3	9	0	1	
4	2	0	7	
5	10	0	3	
6	4	1	10	
7	6	0	4	
8	10	1	7	
9	3	2	3	

Likely to Change Occupation	
0	0
1	0
2	0
3	0
4	1
5	1
6	0
7	0
8	1
9	0

```
[10 rows x 23 columns]
```

	<bound method DataFrame.sum of	Field of Study	Current
Occupation	Age Gender Years of	Experience \	
0	False	False False	False
False			
1	False	False False	False
False			
2	False	False False	False
False			
3	False	False False	False
False			
4	False	False False	False
False			
...	...	...	...
...			

38439	False	False	False	False
False				
38440	False	False	False	False
False				
38441	False	False	False	False
False				
38442	False	False	False	False
False				
38443	False	False	False	False
False				

	Education Level	Industry Growth Rate	Job Satisfaction	\
0	False	False	False	
1	False	False	False	
2	False	False	False	
3	False	False	False	
4	False	False	False	
...	...	...	...	
38439	False	False	False	
38440	False	False	False	
38441	False	False	False	
38442	False	False	False	
38443	False	False	False	

	Work-Life Balance	Job Opportunities	...	Skills Gap	\
0	False	False	...	False	
1	False	False	...	False	
2	False	False	...	False	
3	False	False	...	False	
4	False	False	...	False	
...	...	...	...	...	
38439	False	False	...	False	
38440	False	False	...	False	
38441	False	False	...	False	
38442	False	False	...	False	
38443	False	False	...	False	

	Family Influence	Mentorship Available	Certifications	\
0	False	False	False	
1	False	False	False	
2	False	False	False	
3	False	False	False	
4	False	False	False	
...	...	...	...	
38439	False	False	False	
38440	False	False	False	
38441	False	False	False	
38442	False	False	False	
38443	False	False	False	

Freelancing Experience	Geographic Mobility	Professional Networks \
0	False	False
1	False	False
2	False	False
3	False	False
4	False	False
...	...	...
38439	False	False
38440	False	False
38441	False	False
38442	False	False
38443	False	False
Career Change Events	Technology Adoption	Likely to Change Occupation
0	False	False
1	False	False
2	False	False
3	False	False
4	False	False
...	...	...
38439	False	False
38440	False	False
38441	False	False
38442	False	False
38443	False	False

[38444 rows x 23 columns]>

after handling categorical values:

Field of Study Current Occupation Age Gender Years of
Experience \

0	8	2	28	1
7				
1	5	4	24	1
26				
2	5	1	1	0
27				
3	5	2	13	1
14				
4	0	3	8	0
0				
5	8	5	36	1
39				
6	8	8	31	0
29				
7	8	0	38	0
32				
8	3	8	11	0
38				
9	2	3	10	0
16				

Education Level Industry Growth Rate Job Satisfaction Work-Life
Balance \

0	1	0	6
9			
1	2	1	9
2			
2	2	1	7
2			
3	3	2	6
8			
4	3	1	2
0			
5	0	1	2
0			
6	1	1	6
3			
7	1	0	8
3			
8	0	1	0
4			
9	2	2	7
4			

Job Opportunities ... Skills Gap Family Influence Mentorship

Available \				
0	82	...	7	0
0				
1	54	...	2	2
0				
2	77	...	3	1
0				
3	61	...	1	2
1				
4	7	...	4	1
0				
5	98	...	9	2
1				
6	12	...	5	0
1				
7	76	...	7	2
0				
8	55	...	5	2
0				
9	97	...	5	0
0				
Certifications Freelancing Experience Geographic Mobility \				
0	0		0	1
1	0		1	1
2	0		0	0
3	0		0	0
4	0		1	0
5	0		0	0
6	1		0	1
7	0		0	0
8	1		0	0
9	0		0	0
Professional Networks Career Change Events Technology Adoption \				
0		1	0	0
1		1	1	8
2		1	1	1
3		8	0	0
4		1	0	6
5		9	0	2
6		3	1	9
7		5	0	3
8		9	1	6
9		2	2	2
Likely to Change Occupation				
0			0	
1			0	
2			0	

3	0
4	1
5	1
6	0
7	0
8	1
9	0

[10 rows x 23 columns]

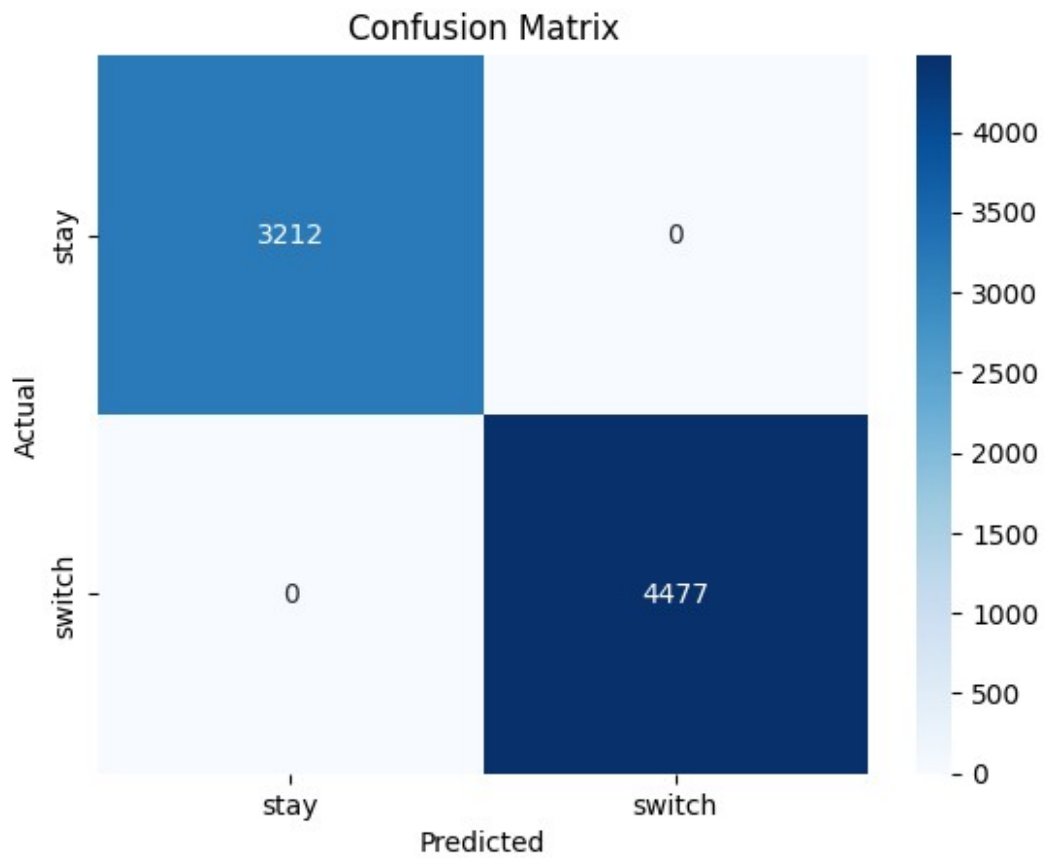
classification report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	3212
1	1.00	1.00	1.00	4477
accuracy			1.00	7689
macro avg	1.00	1.00	1.00	7689
weighted avg	1.00	1.00	1.00	7689

accuracy score:

1.0

confusion matrix:



	principal component 1	principal component 2	Likely to Change
Occupation			
0	16915.051545	32.927938	
0			
1	-3718.948053	4.649542	
0			
2	-10056.948497	27.530931	
0			
3	-6007.948189	11.584410	
0			
4	-14515.947433	-42.565862	
1			

2 component PCA

